Email Security Analysis & Phishing Investigation Report

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Internship: Future Interns – Cyber Security Internship

Project Title: Email Security Analysis & Phishing Investigation Report

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Analysis Environment: Kali Linux on Oracle VirtualBox

Assessment Type: Email Header Forensics, Authentication Validation, Domain Intelligence, DNS Analysis, Social Engineering Detection

Email Security Analysis & Phishing Investigation

INTRODUCTION

This report presents a detailed forensic assessment of suspicious and potentially malicious email samples analyzed in a controlled cybersecurity environment. The purpose of this investigation is to identify phishing characteristics, validate email authenticity, detect infrastructure anomalies, and evaluate overall security risk associated with each message.

Phishing emails remain one of the most common cyberattack vectors, where threat actors impersonate trusted organizations to manipulate users into revealing sensitive information, clicking malicious links, or interacting with fraudulent infrastructure. The objective of this investigation was to determine whether the messages represented legitimate communication, suspicious activity, or active phishing threats.

The investigation combined email header forensics, authentication analysis, DNS intelligence, domain verification, sender reputation analysis, and behavioral threat profiling. Each sample was examined using industry-standard investigative methodology similar to Security Operations Center (SOC) workflows.

OBJECTIVE

The primary objectives of this investigation are:

- Identify phishing indicators in real-world email samples
- Analyze email headers for authenticity verification
- Detect spoofed domains and misleading sender identities
- Investigate suspicious links and domain infrastructure
- Understand social engineering tactics used by attackers
- Document findings using professional cyber forensic reporting standards

TOOLS USED

Security Analysis Stack

- Kali Linux
- grep
- dig
- whois
- Google Admin Toolbox
- MXToolbox
- Raw .eml Header Analysis

METHODOLOGY

The investigation followed a structured workflow similar to Security Operations Center (SOC) incident analysis.

Investigation Objectives

- Identify phishing indicators in real-world email samples
- Validate sender identity and email authenticity
- Detect spoofed domains and infrastructure anomalies
- Identify social engineering and impersonation tactics
- Document findings using professional cyber forensic reporting standards

Executive Classification Summary

Sample	**Classification**	**Risk Level**
Sample 01	SAFE	Low
Sample 02	PHISHING	Critical
Sample 03	SUSPICIOUS	Medium

ANALYSIS WORKFLOW

Tools Used

- Kali Linux
- grep
- dig
- whois
- Google Admin Toolbox
- MXToolbox
- Raw .eml Header Analysis

Investigation Methodology

Step 1 — Email Collection

Suspicious email samples were collected in .eml format for forensic analysis.

Step 2 — Header Extraction

Critical header fields including From, Return-Path, Message-ID, and Received entries were extracted.

Step 3 — Authentication Validation

SPF, DKIM, and DMARC records were validated using header analysis tools.

Step 4 — Infrastructure Intelligence

Domain ownership, DNS resolution, and mail routing behavior were validated using WHOIS and dig.

Step 5 — Behavioral Analysis

Message content, urgency triggers, impersonation tactics, and user manipulation indicators were examined.

Investigation Workflow

Email Collection → Header Extraction → Authentication Validation → Domain Intelligence → DNS Validation → Behavioral Analysis → Final Classification

CASE STUDY 01 — X Security Alert

Sender: verify@x.com

Subject: Security alert: new or unusual X login

Technical Findings

- SPF: PASS
- DKIM: PASS
- DMARC: PASS

Behavioral Findings

- Legitimate security notification
- No credential harvesting indicators
- Infrastructure alignment verified

Final Classification

SAFE

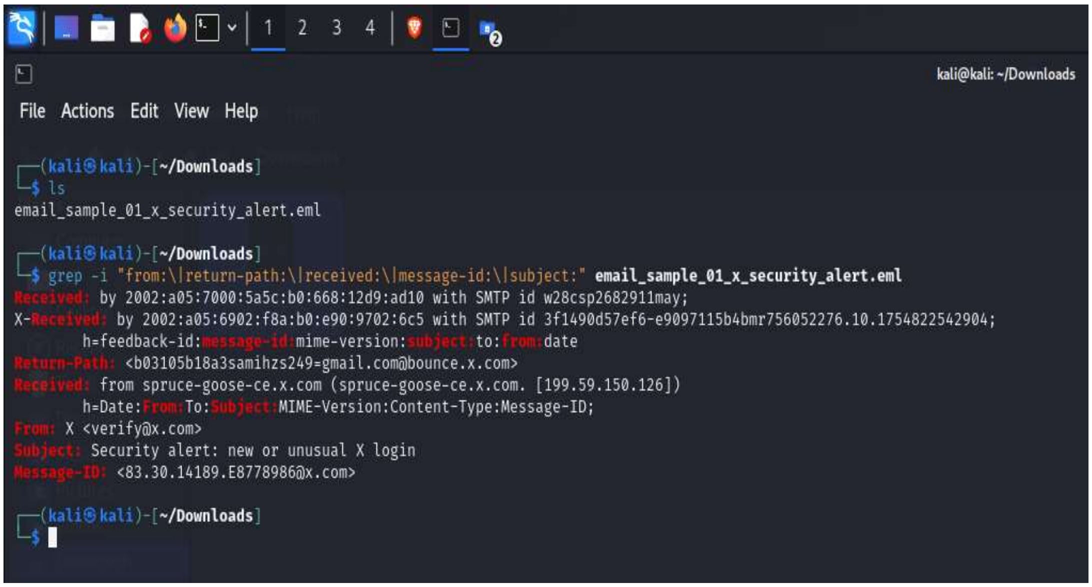
Risk Level

LOW

Evidence Documentation

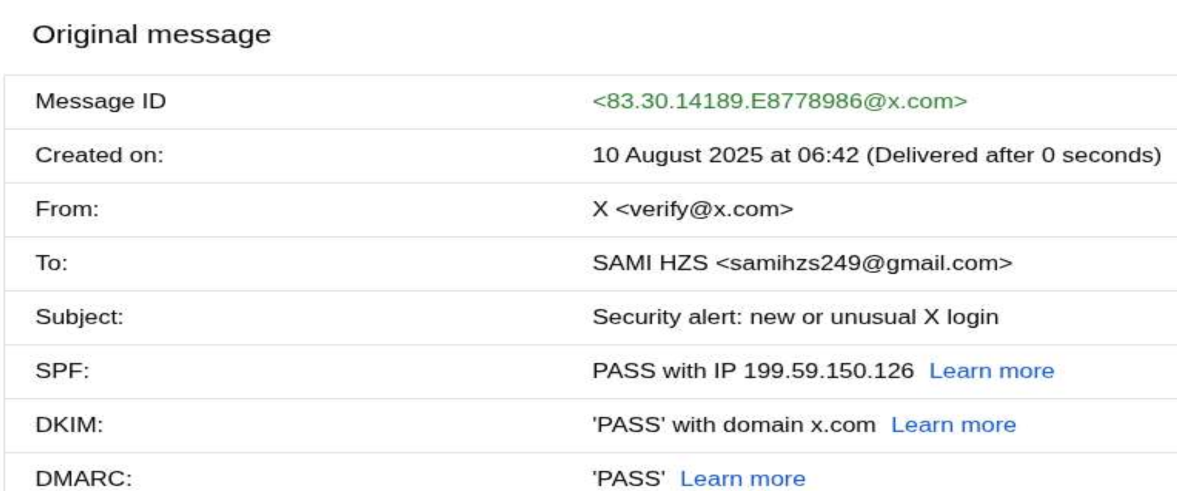
📷 Evidence Block A — Email Header Forensics

This section contains the raw header extraction obtained from the original .eml file. Includes critical fields such as From, Return-Path, Message-ID, and Received headers. This evidence helps establish sender identity, routing path, and header consistency.



📷 Evidence Block B — Authentication Validation

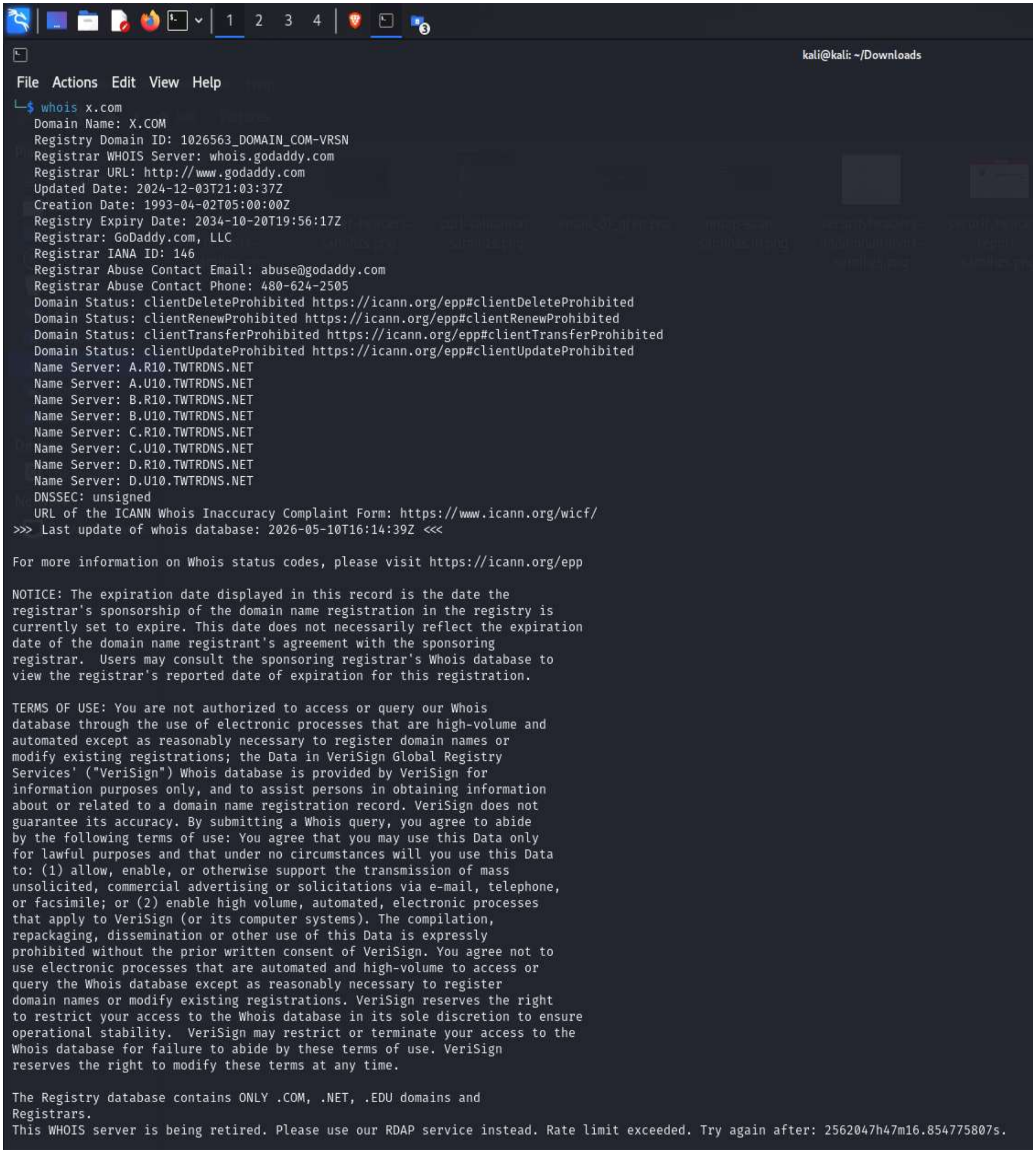
Inserted authentication verification screenshots showing SPF, DKIM, and DMARC validation results. This evidence confirms whether the sending infrastructure passed standard email security checks.


Original message

Message ID	<83.30.14189.E8778986@x.com>
Created on:	10 August 2025 at 06:42 (Delivered after 0 seconds)
From:	X <verify@x.com>
To:	SAMI HZS <samihzs249@gmail.com>
Subject:	Security alert: new or unusual X login
SPF:	PASS with IP 199.59.150.126 Learn more
DKIM:	'PASS' with domain x.com Learn more
DMARC:	'PASS' Learn more

📷 Evidence Block C — Infrastructure Intelligence

This section includes domain investigation evidence from WHOIS, DIG and DNS lookups records. These findings help verify domain ownership, infrastructure legitimacy, and operational trust.



```
File  Actions  Edit  View  Help
kali@kali: ~/Downloads
└─$ whois x.com
   Domain Name: X.COM
   Registry Domain ID: 1026563_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.godaddy.com
   Registrar URL: http://www.godaddy.com
   Updated Date: 2024-12-03T21:03:37Z
   Creation Date: 1993-04-02T05:00:00Z
   Registry Expiry Date: 2034-10-20T19:56:17Z
   Registrar: GoDaddy.com, LLC
   Registrar IANA ID: 146
   Registrar Abuse Contact Email: abuse@godaddy.com
   Registrar Abuse Contact Phone: 480-624-2505
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Name Server: A.R10.TWTRDNS.NET
   Name Server: A.U10.TWTRDNS.NET
   Name Server: B.R10.TWTRDNS.NET
   Name Server: B.U10.TWTRDNS.NET
   Name Server: C.R10.TWTRDNS.NET
   Name Server: C.U10.TWTRDNS.NET
   Name Server: D.R10.TWTRDNS.NET
   Name Server: D.U10.TWTRDNS.NET
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-05-10T16:14:39Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar.  Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
operational stability.  VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.

The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
This WHOIS server is being retired. Please use our RDAP service instead. Rate limit exceeded. Try again after: 2562047h47m16.854775807s.
```

```
┌──(kali㉿kali)-[~/Downloads]
└─$ dig x.com | head -25

; <<>> DiG 9.19.19-1-Debian <<>> x.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 33031
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;x.com.                         IN      A

;; ANSWER SECTION:
x.com.                  116     IN      A       172.66.0.227

;; Query time: 8 msec
;; SERVER: 192.168.29.1#53(192.168.29.1) (UDP)
;; WHEN: Sun May 10 14:44:21 EDT 2026
;; MSG SIZE  rcvd: 39
```

📷 **Evidence Block D — Threat Verification**

Includes supporting validation from Google Header Analyzer This evidence supports final threat classification.

≡ Google Admin Toolbox Messageheader

MessageId	83.30.14189.E8778986@x.com
Created at:	8/10/2025, 6:42:22 AM EDT (Delivered after 1 sec)
From:	X <verify@x.com>
To:	SAMI HZS <samihzs249@gmail.com>
Subject:	Security alert: new or unusual X login
DKIM:	**pass** with domain x.com Learn more

#	Delay	From *		To *	Protocol	Time received
0		spruce-goose-ce.x.com.	→ [Google]	mx.google.com	ESMTPS	8/10/2025, 6:42:22 AM EDT
1			→ [Google]	2002:a05:6902:f8a:b0:e90:9702:6c5	SMTP	8/10/2025, 6:42:22 AM EDT
2	1 sec		→ [Google]	2002:a05:7000:5a5c:b0:668:12d9:ad10	SMTP	8/10/2025, 6:42:23 AM EDT

Analyst Interpretation

The collected evidence confirms that the sender infrastructure, authentication records, and behavioral indicators align with the final classification of this email sample. Technical validation combined with contextual analysis reduced the possibility of false positives and supported accurate threat assessment.

CASE STUDY 02 — UPS Brand Impersonation

Sender: UPS Partner Hoffman_John_49321@ali001.sarakzit.za.com

Subject: Exclusive UPS Partnership Opportunity - Unlock Your Earnings!

Technical Findings

- SPF: PASS
- DKIM: PASS
- DMARC: NONE
- DNS Result: NXDOMAIN

Behavioral Findings

- UPS brand impersonation
- Financial lure strategy detected
- Sender domain mismatch
- Social engineering indicators present

Attack Flow

Brand Trust → Financial Incentive → User Manipulation

Final Classification

PHISHING

Risk Level

HIGH

Evidence Documentation

📷 Evidence Block A — Email Header Forensics

This section contains the raw header extraction obtained from the original .eml file. Include critical fields such as From, Return-Path, Message-ID, and Received headers. This evidence helps establish sender identity, routing path, and header consistency.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ grep -i "from:\|return-path:\|received:\|message-id:\|subject:" sample-3864.eml
Received: from SA1P223MB0791.NAMP223.PROD.OUTLOOK.COM (::1) by
 h=From:Date:Subject:Message-ID:Content-Type:MIME-Version:X-MS-Exchange-AntiSpam-MessageData-ChunkCount:X-MS-Exchange-AntiSpam-MessageData-0:X-MS-Exchange-AntiSpam-MessageData-1;
Received: from PH7PR17CA0007.namprd17.prod.outlook.com (2603:10b6:510:324::18)
Received: from SN1PEPF000252A2.namprd05.prod.outlook.com
Received: from APC01-PSA-obe.outbound.protection.outlook.com (52.100.0.236) by
 OriginalChecksum:8B7083812E5B481C3419DAE7BA97E3FA709AED6D0347BBA60C243EC668033D44;UpperCasedChecksum:4F917DE21E7C2FC2B92624D4EFAA785C664CCB0475BA6B6283873F8C47B01721;SizeAsReceived:30980;Count:44
 h=From:Date:Subject:Message-ID:Content-Type:MIME-Version:X-MS-Exchange-AntiSpam-MessageData-ChunkCount:X-MS-Exchange-AntiSpam-MessageData-0:X-MS-Exchange-AntiSpam-MessageData-1;
 h=From:Date:Subject:Message-ID:Content-Type:MIME-Version:X-MS-Exchange-SenderADCheck;
From: UPS Partner <Hoffman_John_49321@ali001.sarakzit.za.com >
Subject: "Exclusive UPS Partnership Opportunity - Unlock Your Earnings!"
Message-ID: <ZF3lf1bGTw-Ct-Y9ADdP_oRwS.P5LqQmTHg-__Ip@z2yyC1ExTh0IOHNsB61tveiBpfZxNHujI5BuousKcXkrDsVLvUfIU837cF6wP0KWIxJWk2lBk9CLkVtcO0JEgwb.dcccd.edu>
Return-Path: Hoffman_John_49321@ali001.sarakzit.za.com
X-MS-Exchange-Organization-Network-Message-Id:
X-MS-Exchange-CrossTenant-Network-Message-Id: f1a0a464-f736-43b0-cbf3-08dcc91db47b
```

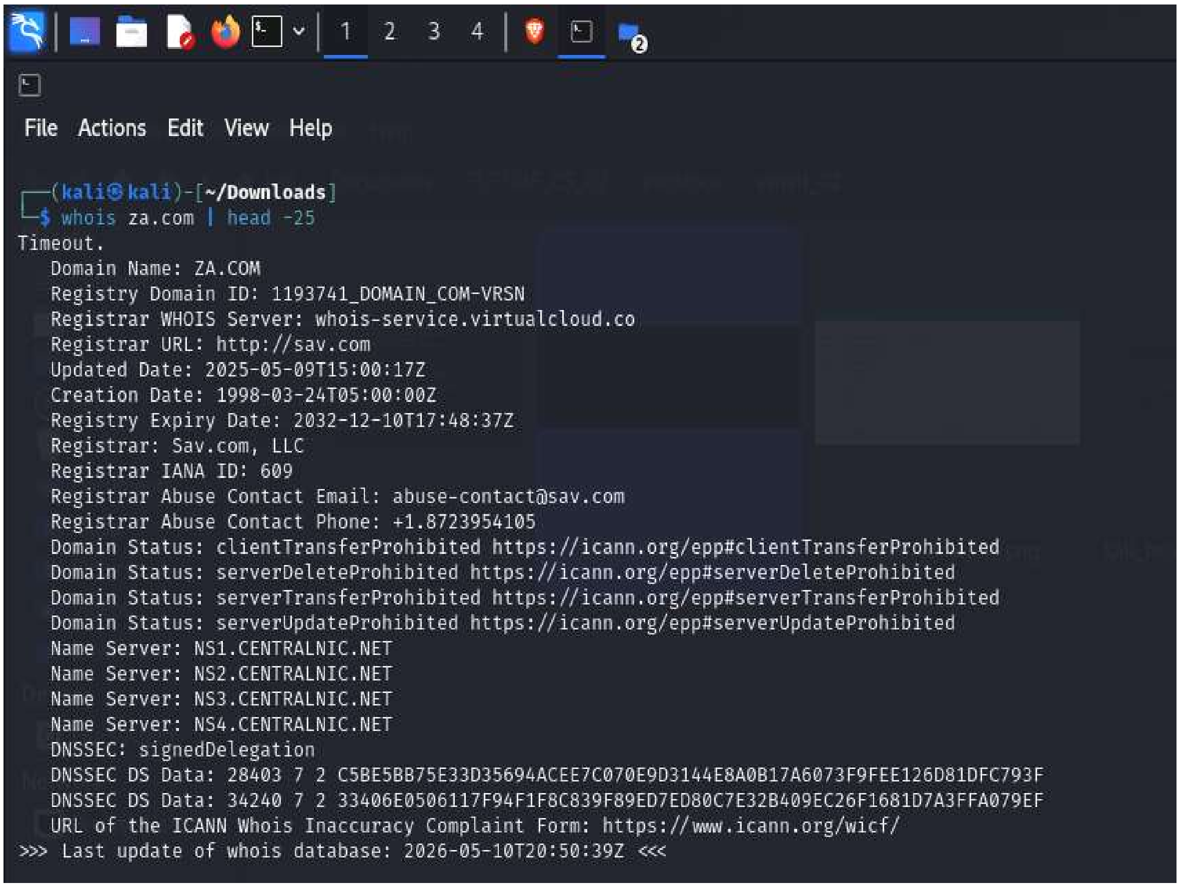
📷 Evidence Block B — Authentication Validation

Inserted authentication verification screenshots showing SPF, DKIM, and DMARC validation results. This evidence confirms whether the sending infrastructure passed standard email security checks.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ grep -i "spf\|dkim\|dmarc\|authentication-results" sample-3864.eml
ARC-Authentication-Results: i=2; mx.microsoft.com 1; spf=pass (sender ip is
 smtp.mailfrom=ali001.sarakzit.za.com; dmarc=none action=none
 header.from=ali001.sarakzit.za.com; dkim=pass (signature was verified)
 spf=[1,1,smtp.mailfrom=ali001.sarakzit.za.com]
 dkim=[1,1,header.d=ali001.sarakzit.za.com]
 dmarc=[1,1,header.from=ali001.sarakzit.za.com])
Authentication-Results: spf=pass (sender IP is 52.100.0.236)
 smtp.mailfrom=ali001.sarakzit.za.com; dkim=pass (signature was verified)
 header.d=ali001.sarakzit.za.com;dmarc=none action=none
Received-SPF: Pass (protection.outlook.com: domain of ali001.sarakzit.za.com
ARC-Authentication-Results: i=1; mx.microsoft.com 1; spf=pass
 smtp.mailfrom=ali001.sarakzit.za.com; dmarc=pass action=none
 header.from=ali001.sarakzit.za.com; dkim=pass
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed;
Authentication-Results-Original: dkim=none (message not signed)
 header.d=none;dmarc=none action=none header.from=ali001.sarakzit.za.com;
 =?iso-8859-1?Q?vxfZnvVx8vuKTY3eKyuVSrYed7SCJiFQX8GvGo5atCCsPFHWYR62CQxGsD?=
```

📷 Evidence Block C — Infrastructure Intelligence

This section includes domain investigation evidence from WHOIS, DIG and DNS lookups records. These findings help verify domain ownership, infrastructure legitimacy, and operational trust.

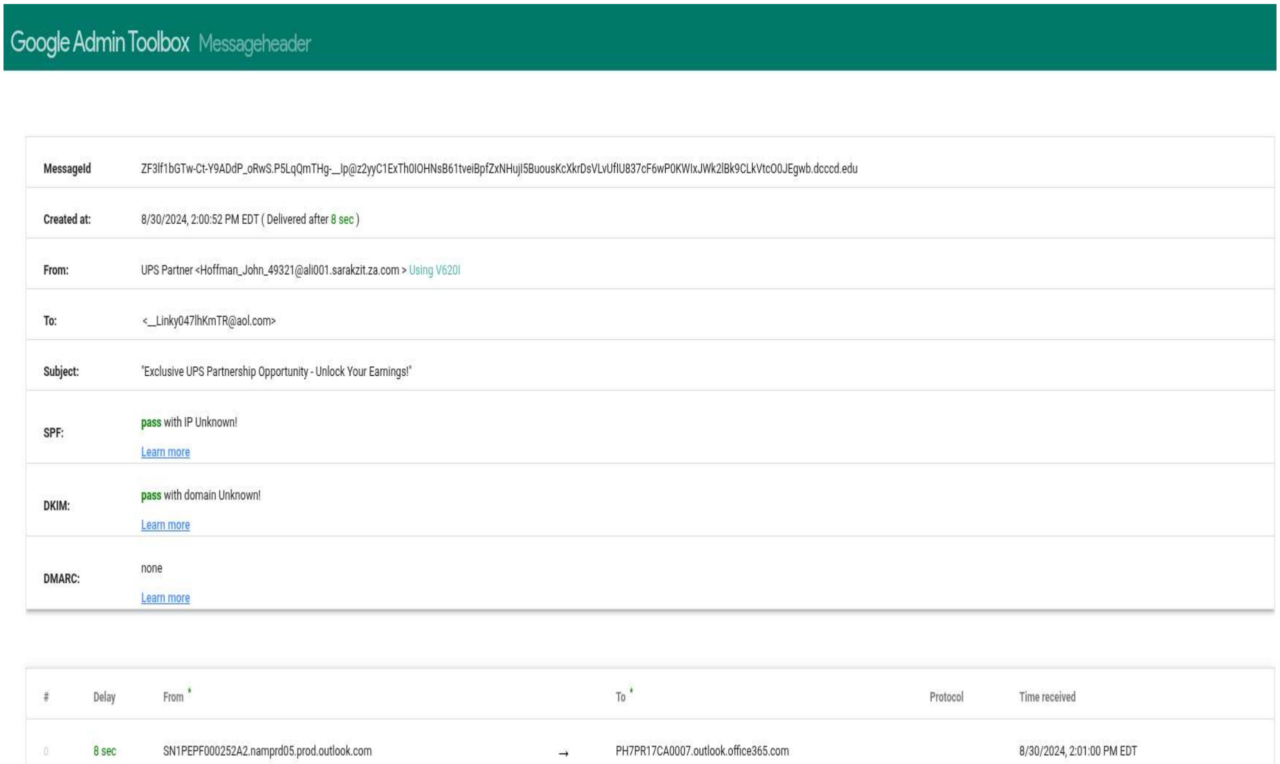


```
File  Actions  Edit  View  Help

┌──(kali㉿kali)-[~/Downloads]
└─$ whois za.com | head -25
Timeout.
   Domain Name: ZA.COM
   Registry Domain ID: 1193741_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois-service.virtualcloud.co
   Registrar URL: http://sav.com
   Updated Date: 2025-05-09T15:00:17Z
   Creation Date: 1998-03-24T05:00:00Z
   Registry Expiry Date: 2032-12-10T17:48:37Z
   Registrar: Sav.com, LLC
   Registrar IANA ID: 609
   Registrar Abuse Contact Email: abuse-contact@sav.com
   Registrar Abuse Contact Phone: +1.8723954105
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
   Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
   Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
   Name Server: NS1.CENTRALNIC.NET
   Name Server: NS2.CENTRALNIC.NET
   Name Server: NS3.CENTRALNIC.NET
   Name Server: NS4.CENTRALNIC.NET
   DNSSEC: signedDelegation
   DNSSEC DS Data: 28403 7 2 C5BE5BB75E33D35694ACEE7C070E9D3144E8A0B17A6073F9FEE126D81DFC793F
   DNSSEC DS Data: 34240 7 2 33406E0506117F94F1F8C839F89ED7ED80C7E32B409EC26F1681D7A3FFA079EF
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-05-10T20:50:39Z <<<
```

```
┌──(kali㉿kali)-[~/Downloads]
└─$ dig ali001.sarakzit.za.com

; <<>> DiG 9.19.19-1-Debian <<>> ali001.sarakzit.za.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 15184
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1280
;; QUESTION SECTION:
;ali001.sarakzit.za.com.            IN      A

;; AUTHORITY SECTION:
za.com.             3600    IN      SOA     ns0.centralnic.net. hostmaster.centralnic.net. 3000565969 900 1800 6048000 3600

;; Query time: 592 msec
;; SERVER: 192.168.29.1#53(192.168.29.1) (UDP)
;; WHEN: Sun May 10 16:56:30 EDT 2026
;; MSG SIZE  rcvd: 116
```

📷 Evidence Block D — Threat Verification

Includes supporting validation from Google Header Analyzer This evidence supports final threat classification.



Analyst Interpretation

The collected evidence confirms that the sender infrastructure, authentication records, and behavioral indicators align with the final classification of this email sample. Technical validation combined with contextual analysis reduced the possibility of false positives and supported accurate threat assessment.

CASE STUDY 03 — Contextual Mismatch Email

Sender: deaglefilomena@richmondanglicanchurch.com

Subject: Stay Ahead: The Future of Energy is Here

Technical Findings

- SPF: NONE
- DKIM: PASS
- DMARC: NONE

Behavioral Findings

- Domain context mismatch identified
- FOMO persuasion tactics observed
- Partial technical legitimacy

Final Classification

SUSPICIOUS

Risk Level

MEDIUM

Confidence Score

4/5

Evidence Documentation

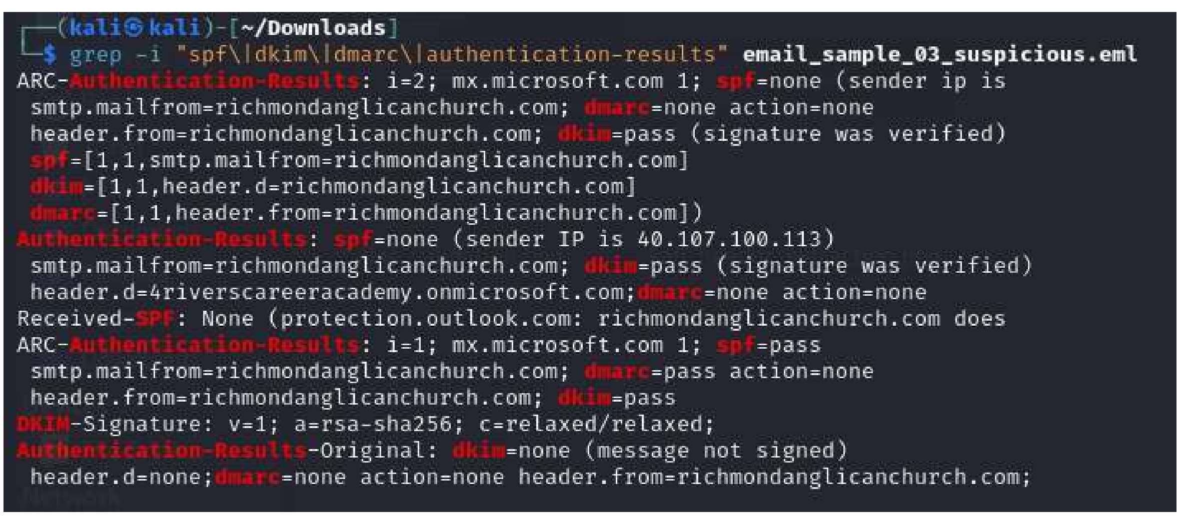
📷 **Evidence Block A — Email Header Forensics**

This section contains the raw header extraction obtained from the original .eml file. Includes critical fields such as From, Return-Path, Message-ID, and Received headers. This evidence helps establish sender identity, routing path, and header consistency.

```
kali@kali: ~/Downloads

┌──(kali㉿kali)-[~]
└─$ cd Downloads

┌──(kali㉿kali)-[~/Downloads]
└─$ grep -i "from:\|return-path:\|received:\|message-id:\|subject:" email_sample_03_suspicious.eml
Received: from SN7PR19MB7451.namprd19.prod.outlook.com (2603:10b6:806:32b::12)
 h=From:Date:Subject:Message-ID:Content-Type:MIME-Version:X-MS-Exchange-AntiSpam-MessageData-ChunkCount:X-MS-Exchange-AntiSpam-MessageData-0:X-MS-Exchange-AntiSpam-MessageData-1;
Received: from PH8PR15CA0005.namprd15.prod.outlook.com (2603:10b6:510:2d2::12)
Received: from SN1PEPF000252A1.namprd05.prod.outlook.com
Received: from NAM04-BN8-obe.outbound.protection.outlook.com (40.107.100.113)
 OriginalChecksum:C0859458BD14B6225B66DE81CA6424DAEFAFF370C9974B521F2BBEF97CFC1900;UpperCasedChecksum:C80E26BEA4AB831294A0A3989528AD127BB80858047B3D6E0AD8BD18BB846B43;SizeAsReceived:8631;Count:36
 h=From:Date:Subject:Message-ID:Content-Type:MIME-Version:X-MS-Exchange-AntiSpam-MessageData-ChunkCount:X-MS-Exchange-AntiSpam-MessageData-0:X-MS-Exchange-AntiSpam-MessageData-1;
 h=From:Date:Subject:Message-ID:Content-Type:MIME-Version:X-MS-Exchange-SenderADCheck;
Received: from BYAPR03MB4854.namprd03.prod.outlook.com (2603:10b6:a03:135::31)
Received: from BYAPR03MB4854.namprd03.prod.outlook.com
From: "=?utf-8?B?RWNvIFNhdmluZ3PigK8gLSBFbG9uIE11c2s=?="
Subject: Stay Ahead: The Future of Energy is Here
Return-Path: deaglefilomena@richmondanglicanchurch.com
Message-ID:
X-MS-Exchange-Organization-Network-Message-Id:
X-MS-Exchange-CrossTenant-Network-Message-Id: a62ddb21-d395-40bb-555d-08dbc381ed93
```

📷 Evidence Block B — Authentication Validation

Insert authentication verification screenshots showing SPF, DKIM, and DMARC validation results. This evidence confirms whether the sending infrastructure passed standard email security checks.



```
┌──(kali㉿kali)-[~/Downloads]
└─$ grep -i "spf\|dkim\|dmarc\|authentication-results" email_sample_03_suspicious.eml
ARC-Authentication-Results: i=2; mx.microsoft.com 1; spf=none (sender ip is
 smtp.mailfrom=richmondanglicanchurch.com; dmarc=none action=none
 header.from=richmondanglicanchurch.com; dkim=pass (signature was verified)
 spf=[1,1,smtp.mailfrom=richmondanglicanchurch.com]
 dkim=[1,1,header.d=richmondanglicanchurch.com]
 dmarc=[1,1,header.from=richmondanglicanchurch.com])
Authentication-Results: spf=none (sender IP is 40.107.100.113)
 smtp.mailfrom=richmondanglicanchurch.com; dkim=pass (signature was verified)
 header.d=4riverscareeracademy.onmicrosoft.com;dmarc=none action=none
Received-SPF: None (protection.outlook.com: richmondanglicanchurch.com does
ARC-Authentication-Results: i=1; mx.microsoft.com 1; spf=pass
 smtp.mailfrom=richmondanglicanchurch.com; dmarc=pass action=none
 header.from=richmondanglicanchurch.com; dkim=pass
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed;
Authentication-Results-Original: dkim=none (message not signed)
 header.d=none;dmarc=none action=none header.from=richmondanglicanchurch.com;
```

📷 **Evidence Block C — Infrastructure Intelligence**

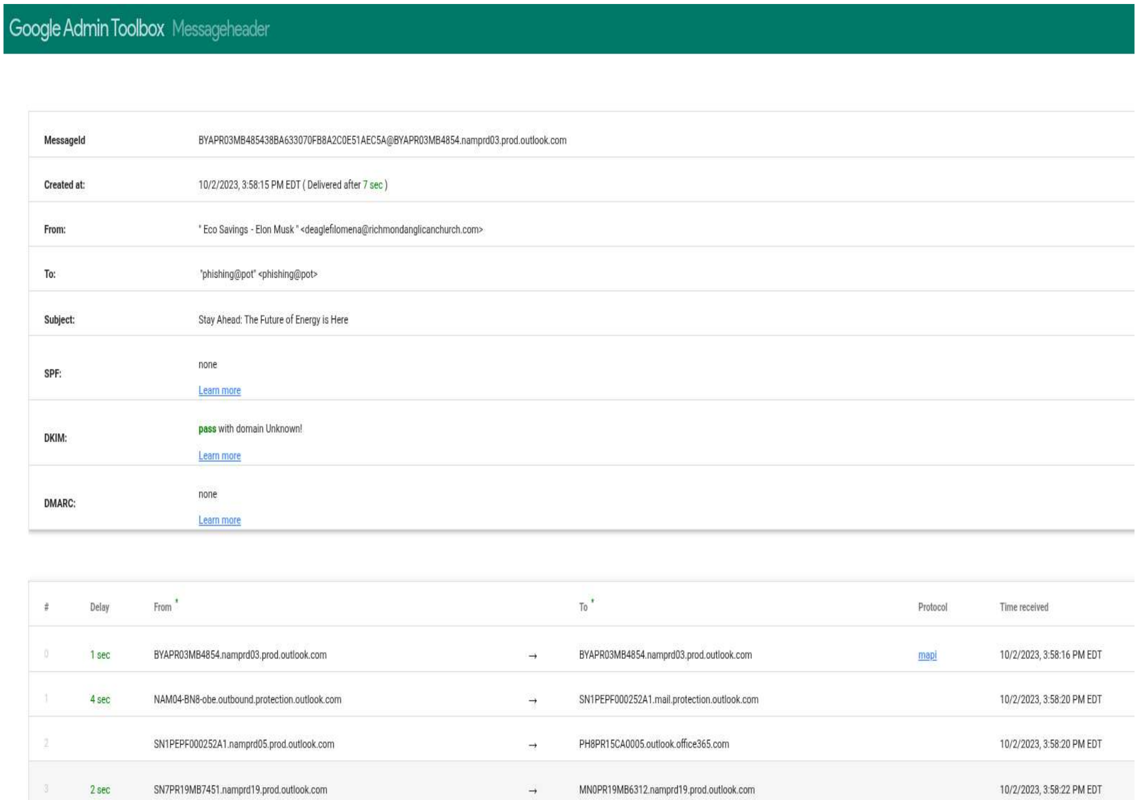
This section includes domain investigation evidence from WHOIS, DIG and DNS lookups records. These findings help verify domain ownership, infrastructure legitimacy, and operational trust.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ whois richmondanglicanchurch.com | head -25
getaddrinfo(www.gname.com/whois): Name or service not known
   Domain Name: RICHMONDANGLICANCHURCH.COM
   Registry Domain ID: 2986413210_DOMAIN_COM-VRSN
   Registrar WHOIS Server: www.gname.com/whois
   Registrar URL: http://www.gname.com
   Updated Date: 2026-04-20T16:30:43Z
   Creation Date: 2025-05-25T18:28:43Z
   Registry Expiry Date: 2026-05-25T18:28:43Z
   Registrar: Gname 205 Inc
   Registrar IANA ID: 4218
   Registrar Abuse Contact Email: ZYS@GNAME.COM
   Registrar Abuse Contact Phone: +65.65189986
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Name Server: LOU.NS.CLOUDFLARE.COM
   Name Server: LUCY.NS.CLOUDFLARE.COM
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-05-10T22:11:10Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar.  Users may consult the sponsoring registrar's Whois database to
```

```
┌──(kali㉿kali)-[~/Downloads]
└─$ dig richmondanglicanchurch.com

; <<>> DiG 9.19.19-1-Debian <<>> richmondanglicanchurch.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 60789
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1280
;; QUESTION SECTION:
;richmondanglicanchurch.com.    IN      A

;; AUTHORITY SECTION:
richmondanglicanchurch.com. 1800 IN     SOA     lou.ns.cloudflare.com. dns.cloudflare.com. 2402189615 10000 2400 604800 1800

;; Query time: 1852 msec
;; SERVER: 192.168.29.1#53(192.168.29.1) (UDP)
;; WHEN: Sun May 10 18:15:43 EDT 2026
;; MSG SIZE  rcvd: 113
```

📷 Evidence Block D — Threat Verification

Includes supporting validation from Google Header Analyzer This evidence supports final threat classification.



Google Admin Toolbox Messageheader

MessageId	BYAPR03MB485438BA633070FB8A2C0E51AEC5A@BYAPR03MB4854.namprd03.prod.outlook.com
Created at:	10/2/2023, 3:58:15 PM EDT (Delivered after 7 sec)
From:	" Eco Savings - Elon Musk " <deaglefilomena@richmondanglicanchurch.com>
To:	"phishing@pot" <phishing@pot>
Subject:	Stay Ahead: The Future of Energy is Here
SPF:	none Learn more
DKIM:	pass with domain Unknown! Learn more
DMARC:	none Learn more

#	Delay	From *		To *	Protocol	Time received
0	1 sec	BYAPR03MB4854.namprd03.prod.outlook.com	→	BYAPR03MB4854.namprd03.prod.outlook.com	mapi	10/2/2023, 3:58:16 PM EDT
1	4 sec	NAM04-BN8-obe.outbound.protection.outlook.com	→	SN1PEPF000252A1.mail.protection.outlook.com		10/2/2023, 3:58:20 PM EDT
2		SN1PEPF000252A1.namprd05.prod.outlook.com	→	PH8PR15CA0005.outlook.office365.com		10/2/2023, 3:58:20 PM EDT
3	2 sec	SN7PR19MB7451.namprd19.prod.outlook.com	→	MN0PR19MB6312.namprd19.prod.outlook.com		10/2/2023, 3:58:22 PM EDT

Analyst Interpretation

The collected evidence confirms that the sender infrastructure, authentication records, and behavioral indicators align with the final classification of this email sample. Technical validation combined with contextual analysis reduced the possibility of false positives and supported accurate threat assessment.

COMPARATIVE ANALYSIS

Sample	Authentication	Behavioral Indicators	Infrastructure Trust	Final Verdict
Sample 01	Strong	Legitimate	Trusted	SAFE
Sample 02	Partial	Malicious	Invalid	PHISHING
Sample 03	Partial	Suspicious	Unclear	SUSPICIOUS

INDICATORS OF COMPROMISE (IOCs)

Identified Threat Indicators

- Domain and sender identity mismatches
- Missing or weak DMARC enforcement
- Suspicious or non-resolving sender infrastructure
- Social engineering through urgency, rewards, or fear
- Contextual mismatch between sender identity and email content
- Repeated campaign structures across multiple samples

Threat Intelligence Summary

These indicators demonstrate that attackers increasingly leverage partially legitimate infrastructure to bypass basic filtering mechanisms.

KEY SECURITY LESSONS

Strategic Security Findings

The investigation revealed that modern phishing campaigns increasingly abuse partially legitimate infrastructure to evade basic detection mechanisms. Technical trust indicators alone are insufficient without contextual analysis, sender reputation validation, and behavioral correlation.

Critical Analyst Findings

- SPF and DKIM passing do not guarantee trust
- Domain age alone cannot validate legitimacy
- DNS presence does not confirm business trust
- Context mismatch is a powerful threat indicator
- Social engineering often bypasses technical controls

ORGANIZATIONAL RECOMMENDATIONS

Critical Priority

- Enforce DMARC policy with reject or quarantine mode
- Deploy brand impersonation monitoring across email gateways

High Priority

- Conduct employee phishing simulation and awareness training
- Validate sender domains before engaging with external communications

Medium Priority

- Monitor contextual anomalies in inbound communications
- Review domain intelligence before trusting authentication results

End User Safety Guidelines

Do: Verify sender domains, inspect links before clicking, report suspicious emails.

Do Not: Trust urgency-based requests, download unknown attachments, submit credentials through email links.

RECOMMENDATIONS

Critical

- Enforce DMARC policies
- Monitor brand impersonation attempts

High

- Train employees on phishing detection
- Validate sender infrastructure

Medium

- Investigate contextual anomalies

FINAL CONCLUSION

This project demonstrates practical email threat analysis using authentication verification, domain intelligence, infrastructure validation, and behavioral investigation. The findings reinforce the importance of combining technical evidence with contextual analysis to accurately classify threats.

Signature

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--End of Report--